APPENDIX C - QUICK REFERENCE

PRINT CONSTRAINT: ANSWER_FREE

PRINT CONSTRAINT: GRAYSCALE_SAFE

{{ HANDOUT_TITLE: Essential First Moves, Decision Flowcharts & Verification Cues }}

CAPABILITY 01: OXIDATION STATE

{{ FIRST_MOVE }}:

* Check if element is uncombined => ON = 0.
* Otherwise, write fixed rules above metals:
  Group 1 = +1, Group 2 = +2, Fluorine = -1.
* Check Hydrogen: Non-metal compound = +1,
  metal hydride = -1.
* Check Oxygen: Standard = -2, peroxide = -1.

{{ RULE_OR_DECISION_CUE }}:

Decision Ladder Order:

1. Element alone? => 0
2. Group 1 / 2 metal present? => +1 / +2
3. Fluorine present? => -1
4. Hydrogen? => +1 (or -1 with metal)
5. Oxygen? => -2 (or -1 in peroxide)
6. Sum of all ONs = Net Charge of species.
   Solve algebraically for the unknown.

{{ VERIFICATION_CUE }}:

* Multiply assigned ON by atom count.
* Add all contributions algebraically.
* Sum MUST equal overall molecule/ion charge.
* Check that metal does not exceed max valence.

CAPABILITY 02: AGENT ROLES

{{ FIRST_MOVE }}:

* Connect identical elements from reactants to
  products with an arrow.
* Calculate Delta ON = ON_product - ON_reactant.
* If Delta ON is positive => Element is OXIDISED.
* If Delta ON is negative => Element is REDUCED.

{{ RULE_OR_DECISION_CUE }}:

SELF vs OTHER Decision Matrix:

* What it undergoes (SELF) vs What it acts as (OTHER):
  Undergoes Oxidation => Acts as REDUCING AGENT.
  Undergoes Reduction => Acts as OXIDISING AGENT.

Crucial Role Attachment Rule:

* The agent role belongs to the ENTIRE REACTANT
  species, never an isolated element or product!
  Example: If C in C2O4^2- oxidises, the reducing
  agent is the whole C2O4^2- ion.

{{ VERIFICATION_CUE }}:

* Total electrons lost by reducing agent MUST
  equal total electrons gained by oxidising agent.

CAPABILITY 03: PHYSICAL MODEL GATE

{{ FIRST_MOVE }}:

* Draw 1D axis and choose positive direction (+).
* Translate keywords:
  'From rest' => u = 0.
  'Stops / Braking' => v = 0.
  'Highest point' => v_top = 0.

{{ RULE_OR_DECISION_CUE }}:

Model Gate Decision Tree:

* Constant Velocity? (a = 0) => Use s = vt.
* Constant Acceleration? (a != 0, const):
  1. If time 't' is unknown and not needed:
     Use v^2 = u^2 + 2as.
  2. If final velocity 'v' is not needed:
     Use s = ut + 1/2 at^2.
  3. If displacement 's' is not needed:
     Use v = u + at.
* Free Fall? Set a = -g (if up is positive).

{{ VERIFICATION_CUE }}:

* Units check: [m/s], [m/s2], [m], [s].
* Direction check: Does displacement sign match
  the final coordinate relative to start?